

BANK LOAN REPORT

Comprehensive Technical & Operational Documentation

Author Architecture: Tableau Metadata Module & SQL Portfolio Database Engine

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1. Executive Project Overview

This document compiles the complete end-to-end framework of the **Bank Loan Analysis Platform**. The solution links transactional operational records via an upstream relational database directly with a specialized three-layer interactive data visualization product built inside Tableau.

2. Tableau Business Intelligence Layer Architecture

The visual analytics engine uses metadata calculation rules to compute critical financial tracking figures on the fly. It utilizes built-in timeframe shifting macros to evaluate momentum fields dynamically.

Tableau Measure / Field Name	Logical Formula Pattern	Business Purpose
Total Loan Applications	COUNT([id])	Tracks gross pipeline application volumes across all product segments.
Total Funded Amount	SUM([loan_amount])	Monitors total capital outflows and liquidity allocations.
Total Amount Received	SUM([total_payment])	Tracks direct recovery metrics against funded principal bases.
Avg Interest Rate	AVG([int_rate])	Measures the macro yield yield and revenue profile over variants.
Good vs Bad Loan	Conditional binning isolating 'Charged Off' vs. stable states ('Current', 'Fully Paid').	Segments portfolio risk streams automatically for early threat mitigation.

3. Relational Database Implementation (SQL Scripts)

To ensure seamless cross-platform consistency, the accompanying database environment mirrors the Tableau metrics. Below are the verified analytical scripts utilized for transactional parsing:

KPI Metrics Extraction (Total Applications, MTD, and PMTD Monitoring)

```
SELECT COUNT(id) AS Total_Loan_Applications FROM Bank_Loan_Data;

-- Month-To-Date (MTD) vs Prior Month-To-Date (PMTD) Volume Tracking
SELECT COUNT(id) AS MTD_Total_Loan_Applications FROM Bank_Loan_Data WHERE MONTH(issue_date) = 12 AND Y
EAR(issue_date) = 2021;
SELECT COUNT(id) AS PMTD_Total_Loan_Applications FROM Bank_Loan_Data WHERE MONTH(issue_date) = 11 AND
YEAR(issue_date) = 2021;
```

Portfolio Quality Segmentation Rules (Good vs. Bad Loans Tracking)

```
-- Good Loans Assessment (Active current or completely paid accounts)
SELECT (COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id END)*100)/COUNT(
id) AS Good_loan_percentage FROM Bank_Loan_Data;

-- Bad Loans Leakage Tracking (Charged Off positions affecting liquidity)
SELECT (SUM(CASE WHEN loan_status = 'Charged Off' THEN 1 ELSE 0 END) * 100.0) / COUNT(id) AS Bad_Loan_
Percentage FROM Bank_Loan_Data;
```

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